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Topic : Advancements in the fields of Computers

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ADVANCEMENT IN THE FIELD OF SUPER COMPUTER DEVELOPMENT AND APPLICATIONS

A supercomputer is a computer with a high level of performance compared to a general-purpose computer. Performance of a supercomputer is measured in floating – point operations per second (FLOPS) instead of million instructions per second (MIPS).

The term 'supercomputing' refers to the processing of massively complex or data-laden problems using the concentrated compute resources of multiple computer systems working in parallel. Supercomputing efficiently solves extremely complex or data-intensive problems by concentrating on the processing power of multiple, parallel computers. Supercomputing involves a system working at the maximum potential performance of any computer, typically measured in Petaflops Sample use cases include genomics, astronomical calculations, and so forth.

Supercomputers play an important role in the field of computational science, and are used for a wide range of computationally intensive tasks in various fields, including quantum mechanics, weather forecasting, climate research, oil and gas exploration, molecular modeling (computing the structures and properties of chemical compounds, biological macromolecules, polymers, and crystals), and physical simulations (such as simulations of the early moments of the universe, airplane and spacecraft aerodynamics, the detonation of nuclear weapons, and nuclear fusion).

DEVELOPMENT AND TRENDS

Supercomputers were introduced in the 1960s, and for several decades the fastest was made by Seymour Cray at Control Data Corporation (CDC), Cray Research and subsequent companies bearing his name or monogram.

The US has long been a leader in the supercomputer field, first through Cray's almost uninterrupted dominance of the field, and later through a variety of technology companies. Japan made major strides in the field in the 1980s and 90s, but since then China has become increasingly active in the field. As of June 2018, the fastest

supercomputer on the TOP500 supercomputer list (a respected international tracker of the world's fastest supercomputers) is the Summit, in the United States, with a LINPACK benchmark score of 122.3 PFLOPS, exceeding the previous record holder, Surway TaihuLight, by around 29 PFLOPS. Sunnway TaihuLight's is notable for its use of indigenous chips and is the first Chinese computer to enter to TOP500 list without using handware from the United States. As of June 2018, China had more computers (206) on the TOP500 list than the United States (124); however, US built computers held eight of the top 20 positions, the U.S. has six of the top 10 and China has two.

India's supercomputer program was started in the late 1980s because Cray supercomputers (from the USA) were denied for import due to an arms ambargo imposed on India, as it was a dual-use technology and could be used for developing nuclear weapons.

Supercomputer PARAM 8000 (made by the Centre for Development of Advanced Computing (C-DAC)) was launched on July 1, 1991, is considered India's first supercomputer. It was indigenously built in 1991 by C-DAC and was replicated and installed at ICAD Moscow in 1991 under Russian collaboration.

As of June 2018, India has five systems on the TOP500 list ranking.

Rank	**Site**	**Name**	**Rmax (TFlop/s)**	**Repeak (TFlop/s)**
39	Indian Institute of Tropical Meteorology	Pratyush (Cray – XC40)	3,763.9	4,006.2
66	National Centre for Medium Range Weather Forecasting	Mihir (Cray XC40)	2,570.4	2,808.7
206	Software Company	InC1 – Lenova C1040	1,123.2	1,413.1
327	Indian Institute of Science	SERC – Cray XC40	901.5	1,244.2
496	Indian Institute of Tropical Meteorology	Data Plex DX360M4	719.2	790.7

India's Fastest Supercomputer

- Pratyush, India's fastest supercomputer yet, was unveiled at the Indian Institute of Tropical Meteorology (IITM) in Pune.
- Pratysuh is a collection of several computers that can deliver a peak power of 6.8 petaflops, facilitating the weather forecasting and climate monitoring in the country.
- It will help India with better forecasts regarding monsoon, extreme events, cyclones, tsunamis, earthquakes, air quality, lightning, fishing, hot and cold waves, flood and drought among others. It will enable to map regions in India at a resolution of 3 km and the globe at 12 km.
- **Pratyush is fourth fastest supercomputer in the world** dedicated for weather and climate research. **It follows supercomputing machines in Japan, US and United Kingdom.** It will improve rankings of Indian supercomputer from 300s in Top500 list, a respected international tracker of the world's fastest supercomputers.

FUTURE SUPERCOMPUTERS

The Indian Government has proposed to commit 2.5 billion USD to supercomputing research during the 12th Five-year plan period (2012-2017). The project will be handled by Indian Institute of Science (IISc), Bangalore. Additionally, it was later revealed that India plans to develop a supercomputer with processing power in the exaflop range. C-DAC will develop it within the subsequent five years of approval.

In march 2015, the Indian government approved a seven-year supercomputing program worth $730 million (Rs.4,500 crore). The national Supercomputing grid will consist of 73 geographically-distributing high-performance computing centres linked over a high-speed network. The mission involves both capacity and capability machines and includes standing up three petascale supercomputers.

THREATS TO COMPUTERS

BOTNETS:

What they are? A collection of software robots, or 'bots', that creates an army of infected computers (known as 'zombies') that are remotely controlled by the originator. Yours may be one of them, and you may not even know it.

What they can do?

1. Send spam emails with viruses attached
2. Spread all types of malware
3. Can use your computer as part a denial of service attack against other systems.

HACKING: Hacking is a term used to describe actions taken by someone to gain unauthorizes assess to a computer. They availability of information online on the tools, techniques, and malware makes it easier for even non-technical people to undertake malicious activities.

What it can do?

1. Find weaknesses (or pre-existing bugs) in your security settings and exploit them to access your information.
2. Install a Trojan horse, providing a back door for hackers to enter and search for your information.

MALWARE: Malware is one of the more common ways to infiltrate or damage your computer

What it is? Malicious software that infects your computer, such as computer viruses, worms, Trojan horses, spyware, and adware.

What it can do?

1. Intimidate you with scareware, which is usually a pop-up message that tells you your computer has a security problems or other false information.
2. Reformat the hand drive of your computer causing you to lose all your information.

3. Alter or delete files.
4. Steal sensitive information
5. Send emails on your behalf
6. Take control of your computer and all the software running on it.

PHARMING: Pharming is common type of online fraud.

What it is? A means to point you to malicious and illegitimate website by redirecting the legitimates URL Even if the URL is entered correctly, it can still be redirected to fake website.

What it can do?

Convince you that the site is real and legitimate by spoofing or looking almost identical to the actual site down to the smallest details. You may enter your person information and unknowingly give it to someone with malicious intent.

PHISHING

Phishing is used most often by cyber criminals because. It's easy to execute and can produce the results they're looking for with very little effort.

What it is? Fake emails, text messages and websites created to look like they're from authentic companies. They're sent by criminals to steal personal and financial information from you. This is the known as "spoofing".

What it does?

1. Trick you into gibing them information by asking you to update, validate or confirm your account, it is often presented in a manner than seems official and intimidating, to encourage you to take action.
2. Provides cyber criminals with your username and passwords so that they can access your accounts (your online bank account, shopping accounts, etc) and steal your credit card numbers.

RANSOMWARE

What it is? Ransomware is a type of malware that restricts access to your computer or your files and display a message that demands payment for the restriction to be removed. The two most common means of infection appear to be phishing emails that contain malicious attachments and pop-up website advertisements.

What it can do?

1. There are two common types of ransomware:
   a) **Lockscreen ransomware:** display an image that prevents you from accessing your computer
   b) **Encryption ransomware:** encrypts files on your system's hard drive and sometimes on shared network drives, USB drives, external hard drives, and even some cloud storage drives, preventing you from opening them.
2. Ransomware will display a notification stating that your computer or data have been locked and demanding payment be made for you to regain access. Sometimes the notification states that authorities have detected illegal activity on your computer and that the payment is a fine to avoid prosecution.

SPAM

Spam is one of the more common methods of both sending information out and collecting it from unsuspecting people.

What it is?

The mass distribution of unsolicited messages, advertising or pornography to addresses which can be easily found on the internet through things like social networking sites, company websites and personal blogs.

What it can do?

1. Annoy you with unwanted junk mail.
2. Create a burden for communications service providers and businesses to filter electronic messages.

3. Phish for your information by tricking you into following links or entering details with too-good-to-be-true offers and promotions.
4. Provide a vehicle for malware, scams, fraud and threats to your privacy.

SPOOFING

This technique is often used in conjunction with phishing an attempt to steal your information.

What it is?

A website or email address that is created to look like it comes from a legitimate source. An email address may even include your name, or the name of someone you know, making it difficult to discern site that requires login or other personal information.

SPYWARE

Spyware and adware are often used by third parties to infiltrate your computer.

What it is?

Software that collects personal information about you without you knowing. They often come in the form of a 'free' download and are installed automatically with or without your consent. These are difficult to remove and can infect your computer with viruses.

What it can do?

1. Collect information about you without you knowing about it and give it to third parties.
2. Send your usernames, passwords, surfing habits, list of applications you've downloaded, settings, and even the version of your operating system to third parties.
3. Orange the way your computer runs without your knowledge
4. Take you to unwanted sites or inundate you with uncontrollable pop-up ads.

TROJAN HORSES

What it is? A malicious program that is disguised as, or embedded within, legitimate software. It is an executable file that will install itself and run automatically once it's downloaded.

What it can do?

1. Delete your files
2. Use your computer to hack other computers
3. Watch you through your web cam
4. Log your keystrokes (such as a credit card number you an online purchase.
5. Record usernames, passwords and other personal information.

VIRUSES:

What they are?

Malicious computer programs that are often sent as an email attachment or a download with the intent of infecting your computer, as well as the computers of everyone in your contact list. Just visiting a site can start an automatic download of a virus.

What they can do?

1. Send spam
2. Provide criminals with access to your computer and contact lists.
3. Scan and find personal information like passwords on your computer
4. Hijack your web browser.
5. Disable your security settings.
6. Display unwanted ads.
7. When a program is running, the virus attached to it could infiltrate your hard drive and also spread to USB keys and external hard drives. Any attachment you create using this program and sent to someone else could also infect them with the virus.

WI-FI EAVESDROPPING: WiFi eavesdropping is another method used by cyber criminals to capture personal information.

What it is? Virtual 'listening in' on information that's shared over an unsecured (not encrypted) WiFi network.

What it can do?

1. Potentially access your computer with the right equipment.
2. Steal your personal information including logins and passwords.

WORMS

What they are? A worm, unlike a virus, goes to work on its own without attaching itself to files or programs. It lives in your computer memory, doesn't damage or alter the hard drive and propagates by sending itself to other computers in a network – whether with a company or the internet itself.

What they can do?

1. Spread to everyone in your contact list.
2. Cause a tremendous amount of damage by shutting down parts of the internet, wreaking havoc on an internal network and costing companies enormous amounts of lost revenue.

WPA2 HANDSHAKE VULNERABILITIES

The key reinstallation attack (or Krack) vulnerability allows a malicious actor to read encrypted network traffic on a Wi-FI Protected Access II (WPA2) router and send traffic back to the network.

What it can do?

1. Krack can affect both personal (home users, and small businesses) and enterprise networks. Any devices that are connected to the network, such as laptops, smartphones, smart devices, even an installed USB key, can be read by the attacker. A malicious actor could use this vulnerability to steal sensitive

information, and also insert malware or ransomware that would make a website unsafe to visit.

2. Krack does not divulge Wi-Fi passwords to attackers, nor does it permit a malicious device to be connected to the network. Krack is unable to compromise Virtual Private Networks (VPN) or HTTPS protocols used by online shopping and banking sites.

CYBER WARS

Cyberwarfare is the use or targeting in a battlespace or warfare context of computers, online control systems and networks. It involves both offensive and defensive operations about the threat of cyber-attacks, espionage and sabotage. There has been controversy over whether such operations can be called 'war'. Nevertheless, powers have been developing cyber capabilities and engaged in cyber warfare, both offensively and defensively, including the United States, China, Russia, Israel and the United Kingdom. Two other notable players are Iran and North Korea.

TYPES OF CYBER THREAT

Cyberwarfare can present a multitude of threats to a nation. At the most basic level, cyber-attacks can be used to support traditional warfare. For example, tampering with the operation of air defences via cyber means to facilitate an air attack. Aside from these 'hard' threats, cyber warfare can also contribute to 'soft' threats such as espionage and propagnanda.

Espionage

Espionage is the practice of spying or of using spies, typically by governments to obtain political and military Information. Traditional espionage is not an act of war, nor is cyber-espionage, and both are generally assumed to be ongoing between major powers. Despite this assumption, some incidents can cause serious tensions between nations, and are often described as 'attacks'.

For example Massive spying by the US in many countries, revealed Edward Snowden.

The NSA recording nearly every cell phone conversation in the Bahamas, without the Bahamian government's permission, and similar programs in Kenya, the Philippines, Mexico and Afghanistan.

The 'Titan Rain' probes of American defence contractors computer systems since 2003.

The Office of Personnel Management data breach, in the US, widely attributed to China.

Sabotage

Computers and satellites that coordinate other activities are vulnerable components of a system and could lead to the disruption of equipment. Compromise of military systems, such as C41STAR components that are responsible for orders and communications could lead to their interception or malicious replacement. Power, water, fuel, communications, and transportation infrastructure all may be vulnerable to disruption. According to experts, the civilian realm is also at risk, nothing that the security breaches have already gone beyond stolen credit cared numbers and that potential targets can also include the electric power grid, trains, or the stock market.

In mid July 2010, security experts discovered a malicious software program called Stuxnet that had infiltrated factory computers and had spread to plants around the world. It is considered "the first attack on critical industrial infrastructure that sits at the foundation of modern economies,".

Stuxnet, while extremely effective in delaying Iran's nuclear program for the development of nuclear weaponry, came at a high cost. For the first time, it became clear that not only could cyber weapons be defensive but they could be offensive. The large decentratlization and scale of cyberspace make it extremely difficult to direct from a policy perspective. Non-state actors can play as large a part in the cyberwar space as state actors, which leads to dangerous, sometimes disastrous, consequences. Small groups of highly skilled malware developers can as effectively impact global politics and cyber warfare as warfare as large governmental agencies. A major aspect of this ability lies in the willingness of these groups to share their exploits and developments on the web as a form of arms proliferation. This allows lesser hackers to

become more proficient in creating large scale attacks that once only a small handful was skillful enough to manage. Also, thriving black markets for these kinds of cyber weapons are buying and selling these cyber capabilities to the highest-bidder without regard for consequences.

Denial-of-service attack

In computing, a denial-of-service attack (DoS attack) or distributed denial-of-service attack (DDoS attack) is an attempt to make a machine or network resource unavailable to its intended users. Perpetrators of DoS attacks typically target sites or services hosted on high-profile web servers such as banks, credit card payment gateways, and even root nameservers, DoS attacks may not be limited to computer-based methods, as strategic physical attacks against infrastructure can be just as devastating. For example, cutting undersea communication cables may severely cripple some region and countries with regards to their information warfare ability.

Electrical power grid

The federal government of the United States admits that the electric power grid is susceptible to cyberwarfare. In April 2009, reports surfaced that China and Russia had infiltrated the U.S. electrical grid and left behind software programs that could be used to disrupt the system, according to current and former national security officials. The North American Electric Reliability Corporation (NERC) has issued a public notice that warns that the electrical grid is not adequately protected from cyber attack. China denies intruding into the U.S. electrical grid. One countermeasure would be to disconnect the power grid from the Internet and run the net with droop speed control only. Massive power outages caused by a cyber attack could the economy, district from a simultaneous military attack, or create a national trauma.

On 23 December 2015, what a believed to be a first known successful cyber attack on a power grid tock place in Ukraine leading to temporary blackouts. The cyber attack is attributed to the Russian advanced persistent threat group called 'Sandworm', and it was performed during an ongoing military confrontation.

Propaganda

Propaganda is the deliberate, systematic attempt to shape perceptions, manipulate cognitions, and direct behavior to achieve a response that furthers the desired intent of the propagandist. The internet is a phenomenal means of communication. People can get their message across to a huge audience, and with this opens a window for evil. Terrorist organisations can use this medium to brainwash people.

Cyber propaganda is an effort to control information in whatever form it takes, and influence public opinion. It is a form of psychological warfare, except it uses social media, fake news websites and after digital means. In 2018, Sir Nicholas Carter, Chief of the General Staff of the British Army started that this kind of attack from actors such as Russia "is a form of system warfare that seeks to de-legitimise the political and social system on which our military strength is based".

Economic disruption

In 2017, the WannaCry and Petya cyber attacks, masquerading as ransomware, caused large-scale disruptions in Ukraine as well as to the U.K.'s Nation Health Service, pharmaceutical giant Merck, Maersk shipping company and other organizations around the world. These threats have been recognized as the onset of the fifth generation cyber attacks.

CYBER PEACE AND CYBER COUNTERINTELLIGENCE

The rise of cyber as a warfighting domain has led to efforts to determine how cyberspace can be used to foster peace. For example, the German civil rights panel FifF runs a campaign for cyberspace – for the control of cyberweapons and surveillance technology and against the militarization of cyberspace and the development and stockpiling of offensive exploits and malware. Measures for cyber peace include policymakers developing new rules and norms for warfare, individuals and organizations building new tools and secure infrastructures, promoting open source, the establishment of cybersecurity centers, auditing of critical infrastructure cybersecurity, obligations to disclose vulnerabilities, disarmament, defensive security

strategies, decentralization, education and widely applying relevant tools and infrastructures, encryption and other cyberdefenses.

Cyper counter-intelligence are measures to identify, penetrate, or neutralize foreign operations that use cyber means as the primary tradecraft methodology, as well as foreign intelligence service collection efforts that use traditional methods to gauge cyber capabilities and intentions. One of the hardest issues in cyber counterintelligence is the problem of attribution. Unlike conventional warefare, figuring out who is behind an attack can be very difficult.

Questions:

1. What are the types of Cyber Threats? Discuss.
2. Explain the trends and development of Super Computer and its threats.
3. Give a short note on
   a) BOTNETS
   b) RANSOMWARE
   c) SPYWARE.